BAKERY SALES ANALYSIS

1. Project Overview

Background

The dataset was collected from "The Bread Basket", a bakery located in Edinburgh, Scotland. It contains customer transaction records and the items purchased between October 30, 2016 and April 9, 2017. The original dataset has 20,507 records, over 9,000 transactions, and 4 columns.

This project aims to identify sales patterns and analyze purchasing behavior through transaction-based data analysis. SQL was used for data cleaning and exploratory data analysis (EDA), followed by Microsoft Excel for data visualization.

Dataset Source:
Source: Kaggle – "Bakery Sales Dataset" by Akashdeep Kuila (https://www.kaggle.com/datasets/akashdeepkuila/bakery)

2. Dataset Description

Column	Description
Transaction	Unique identifier for every single transaction
Item	Items purchased
Date_Time	Date and time stamp of the transactions
Period_Day	Part of the day when a transaction is made (morning, afternoon, evening, night)
Weekday_Weekend	Classifies whether a transaction has been made in weekend or weekdays

3. Data Cleaning Process (SQL)

During the data cleaning process, duplicate records were identified and removed, resulting in the deletion of **1,520 duplicate records** based on transaction data. The dataset was also standardized by trimming unnecessary spaces and converting the Date_Time column into a consistent date format.

These steps were performed to improve data quality and ensure more accurate and reliable results during the exploratory data analysis (EDA).

```sql
SELECT *
FROM bakery_sales
WHERE Item = ' ';

SELECT COUNT(*) AS duplicates
FROM (
    SELECT Transaction, Item, date_time, period_day, weekday_weekend
    FROM bakery_sales
    GROUP BY Transaction, Item, date_time, period_day, weekday_weekend
    HAVING COUNT(*) > 1
) d;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

duplicates
1520

```sql
SELECT *
FROM bakery_clean
WHERE count > 1;

DELETE
FROM bakery_clean
WHERE count > 1;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

Transaction	Item	date_time	period_day	weekday_weekend	count

```sql
SELECT Item, TRIM(Item)
FROM bakery_clean;

UPDATE bakery_clean
SET Item = TRIM(Item);

SELECT date_time, TRIM(date_time)
FROM bakery_clean;

UPDATE bakery_clean
SET date_time = TRIM(date_time);

SELECT period_day, TRIM(period_day)
FROM bakery_clean;

UPDATE bakery_clean
SET period_day = TRIM(period_day);

SELECT weekday_weekend, TRIM(weekday_weekend)
FROM bakery_clean;

UPDATE bakery_clean
SET weekday_weekend = TRIM(weekday_weekend);
```

```sql
SELECT *
FROM bakery_clean;

SELECT `date_time`,
STR_TO_DATE(`date_time`, '%m/%d/%Y %H:%i')
FROM bakery_clean;

UPDATE bakery_clean
SET `date_time` = STR_TO_DATE(`date_time`, '%m/%d/%Y %H:%i');

ALTER TABLE bakery_clean
MODIFY COLUMN `date_time` DATETIME;
```

4. Exploratory Data Analysis (EDA)

After the data cleaning process, exploratory data analysis (EDA) was performed to identify sales patterns, customer purchasing behavior, and insights that address the project's business questions. The following questions were used to derive key insights:

1. Which day of the week has more transactions?
2. Which hour of the day is the busiest and least busy?
3. Which items are purchased most frequently?
4. Which item combination(s) are most frequently bought together?
5. Which items are more popular on weekdays compared to weekends?
6. What month has the largest volume of transactions?

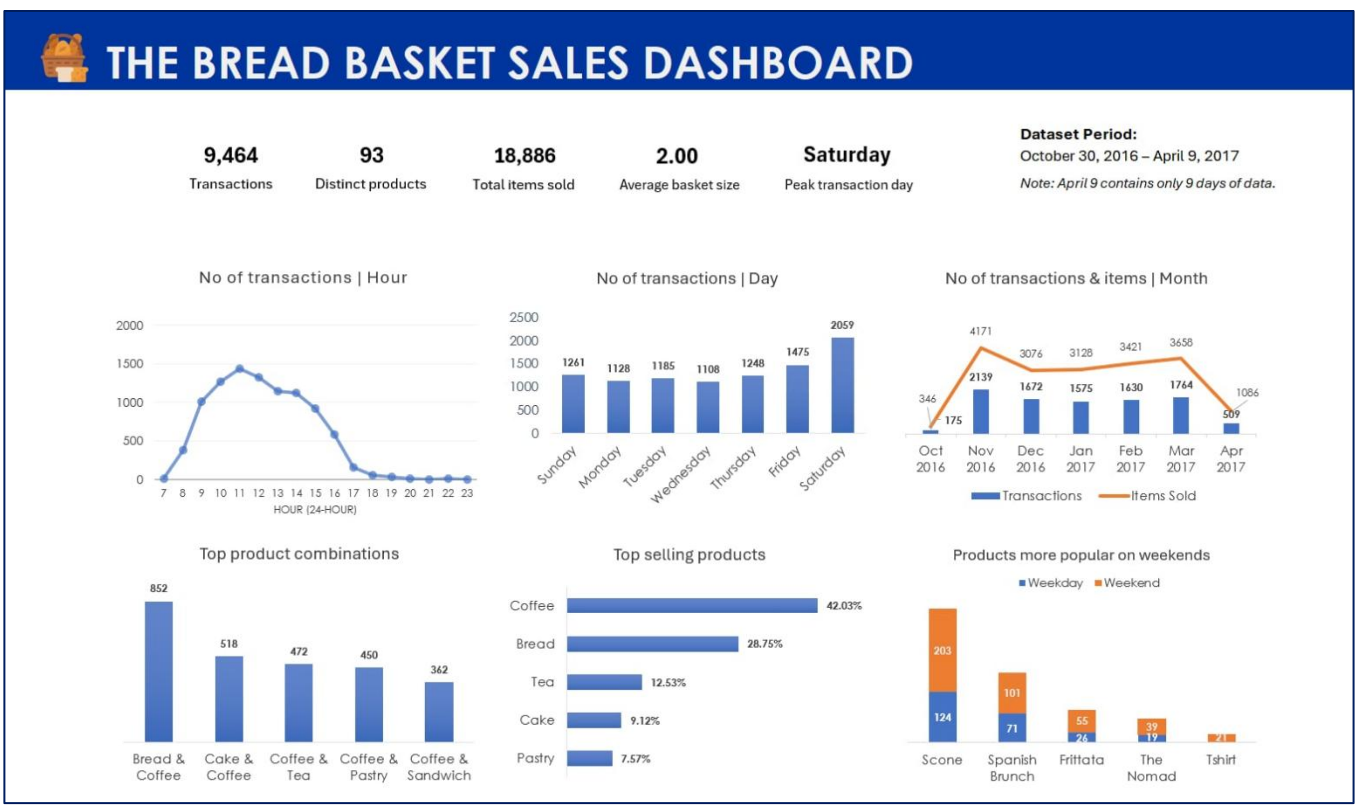
```sql
-- 1. Which period of the day is the busiest? --
SELECT period_day, COUNT(*) AS transaction_count
FROM bakery_clean
GROUP BY period_day
ORDER BY transaction_count DESC;

-- 2. What time of day is the busiest and least busy? --
SELECT HOUR(date_time) AS hour_of_day, COUNT(*) AS transaction_count
FROM bakery_clean
GROUP BY HOUR(date_time)
ORDER BY transaction_count DESC;

SELECT date_time, COUNT(*) AS transaction_count
FROM bakery_clean
GROUP BY date_time
ORDER BY transaction_count ASC;
```

```sql
-- 3. Which items are purchased most frequently? --
SELECT Item, COUNT(*) AS transaction_count
FROM bakery_clean
GROUP BY Item
ORDER BY transaction_count DESC;

-- 4. Which item combination(s) are most frequently bought together? --
-- items purchased together --
SELECT a.Item AS item1, b.Item AS item2, COUNT(*) AS bought_together
FROM bakery_clean a
JOIN bakery_clean b
    ON a.Transaction = b.Transaction
    AND a.Item < b.Item
GROUP BY a.Item, b.Item
ORDER BY bought_together DESC;
```

```sql
-- 5. Which items are more popular on weekdays compared to weekends? --
SELECT Item,
    SUM(CASE WHEN weekday_weekend = 'Weekday' THEN 1 ELSE 0 END) AS weekday_sales,
    SUM(CASE WHEN weekday_weekend = 'Weekend' THEN 1 ELSE 0 END) AS weekend_sales
FROM bakery_clean
GROUP BY Item
HAVING weekend_sales > weekday_sales
ORDER BY weekend_sales - weekday_sales DESC;

-- 6. What month has the largest volume of transactions? --
SELECT MONTH(date_time), COUNT(*) AS monthly_transaction
FROM bakery_clean
GROUP BY MONTH(date_time)
ORDER BY monthly_transaction DESC;
```

5. Dashboard

After completing the exploratory data analysis (EDA), the cleaned dataset was imported into Microsoft Excel for data visualization. PivotTables and charts were used to summarize the analysis results and present the identified sales patterns and customer purchasing behavior in a dashboard. The final dashboard is shown below.



Key Findings:

- From October 30, 2016 to April 9, 2017, the Bread Basket Bakery recorded 9,464 transactions, resulting in 18,886 total items sold across 93 distinct products, with an average basket size of 2.00 items per transaction. This indicates that customers typically purchased two items per visit throughout the analysis period.

- Saturday recorded the highest number of transactions (2,059), making it the busiest day of the week. This suggests that customer visits tend to increase during weekends, likely because more customers have additional free time for dining or shopping.

- The bakery experienced its highest transaction volume at 11:00 AM (1,439 transactions), followed by 10:00 AM and 12:00 PM. This indicates that customer traffic peaks during the late morning to lunchtime period, making it the busiest operating hours of the day.

- November recorded the highest number of transactions (2,139) and items sold (4,171) among all recorded months. Since October contains only two days of data and April includes only nine days, November represents the strongest full month within the available dataset.

- Coffee was the best-selling product throughout the recorded period, accounting for approximately 42% of the total items sold. Bread (29%) ranked second, followed by Tea (13%). This suggests that beverages, particularly coffee, play a significant role in customer purchases and are frequently paired with bakery products.

- Bread and Coffee were the most frequently purchased product combination, with 852 recorded transactions. This suggests that customers commonly purchase beverages alongside bakery items, making the pairing an important sales driver that could be highlighted through meal bundles or promotional offers.

- Several products experienced higher demand during weekends than weekdays. Scone recorded 203 weekend purchases compared to 124 on weekdays, while Spanish Brunch (101 vs. 71) and Frittata (55 vs. 26) followed a similar pattern. These products are commonly associated with breakfast or brunch meals, suggesting that customers tend to choose more leisurely menu items during weekends.

6. Recommendations

- Increase staff allocation during Saturdays and between 10:00 AM and 12:00 PM, as these periods consistently recorded the highest transaction volumes. This may help reduce waiting times and improve customer service during peak hours.

- Maintain adequate inventory levels for Coffee, Bread, and Tea, particularly during peak business hours and weekends, to reduce the risk of stock shortages and ensure product availability.

- Offer bundled meal promotions featuring Coffee with Bread or other frequently paired bakery items. Since Bread and Coffee were the most commonly purchased combination, bundle pricing may encourage additional purchases while improving customer value.

- Increase the availability of weekend-popular products such as Scones, Spanish Brunch, and Frittata. The bakery may also consider introducing additional breakfast or brunch menu items to better match weekend purchasing preferences.

- Introduce limited-time promotions or discounts during off-peak hours and weekdays to encourage customer visits and help balance demand throughout the week.

- Encourage customers to purchase more items per transaction through add-on offers, or checkout promotions. Increasing the average basket size beyond two items could improve overall sales.

7. Conclusion

This project showcases the complete data analysis process, from data cleaning and exploratory data analysis using SQL to data visualization in Microsoft Excel. The insights highlight how transaction data can be used to understand customer behavior and support business decision-making. The project also demonstrates practical skills in data preparation, analysis, visualization, and documentation.